

Chapter 3: Object Oriented Programming Using Python

Chapter Importance (Exam Point of View)

- **1 Long Question** → 8 marks
- **2 Short Questions** → 4 marks
- **2 MCQs** → 2 marks
- **Total** → 14 marks from this chapter

Focus of the chapter:

Fundamentals of Object-Oriented Programming (OOP) using Python

OOP Basic Concepts:

Encapsulation
Inheritance
Polymorphism

3.1 Object-Oriented Programming (OOP) Using Python

Definition: Object-Oriented Programming is a method of designing programs using objects and classes. **Object:** Each object represents a real-world entity with specific properties and behaviors.

Classes: Classes act as templates for creating multiple objects with similar characteristics.

Core Emphasis: OOP emphasizes encapsulation, inheritance, and polymorphism to manage data and behavior efficiently. This approach helps in modeling real-world problems in programming.



Blueprint / Design / Template



Object

Benefits of OOP: Modularity, Reusability, and Easier Maintenance

Modularity: OOP allows code to be divided into modules, making it easier to manage.

Reusability: Classes and objects can be reused in multiple programs or parts of a program, reducing repetition.

Easier Maintenance: Programs built with OOP are easier to maintain and update because changes in one part do not affect the entire system.

Scalability and Adaptability: The structure of OOP supports scalability and adaptability in software development. It also improves collaboration by making code more understandable and organized.

 **Tidbit:** Object-Oriented Programming helps programmers think in terms of real-life objects (like students or cars), making programs easier to understand and manage.

Short Questions

1. What is Object-Oriented Programming (OOP)?
2. What are the main benefits of OOP?
3. What is meant by modularity in OOP?
4. What is reusability in OOP?
5. Why are programs built with OOP easier to maintain?

Long Questions

1. Explain Object-Oriented Programming. Discuss its benefits with examples.
2. Compare procedural programming and Object-Oriented Programming.